

COP4521 Syllabus, Fall 2025

- Location: Mch 201, Time: Tuesday, Thursday 3:05pm-4:20pm
- Instructor: Xin Yuan, (850)6449133, xyuan@fsu.edu
 - Office hours: Tuesday, Thursday 1:15-2:30, ad hoc times announced on canvas, and by appointment
- TAs: TBA

Acknowledgement

- A significant portion of the course materials are adapted from Sharanya Jayaraman and Karen Works's course materials in their previous offerings of this course.

Course Requirements

- Prerequisites

- COP4530: Data Structures, Algorithms, and Generic Programming II
- This also implies that you have a user level knowledge of UNIX: all of our programming assignments should be done and will be graded on the linprog servers.
- Ideally, COP4610 should be a pre-requisite or a co-requisite. Many concepts discussed in the class can be traced back to COP4610.

Website and Textbooks

- Course website: <https://canvas.fsu.edu> or
<http://www.cs.fsu.edu/~xyuan/cop4521>
- Textbooks (optional):
 - *Network Security Essentials: Applications and Standards* (6th Edition) by William Stallings, 2017, ISBN-13: 978013452733
 - *Python Application Programming* by Piyush Kumar, Biswas Parajuli, ISBN-13:9780998169408
 - *Sophomoric Parallelism and Concurrency* by D. Grossman, 2012 (Open Source Textbook)

Course Description

- Over the last decade, information management, security, parallel, and distributed computing have become essential parts of application development.
- In this course students learn how to develop secure, parallel, and distributed applications using Python by properly applying related concepts and principles in database/information management, networking, secure, parallel and distributed computing.

Course Description

- The course touches subjects in the following traditional courses:
 - Python Programming
 - Concurrent, Parallel, and Distributed Programming
 - Networking
 - Security
 - Database
- The pace in this class is very fast. We will move to a completely new topic every 2-3 weeks throughout the semester.

Course Objectives

- **CLO 1:** Develop multiuser applications using Python.
- **CLO 2:** Explain information management and security computing principles such as basic computer security concepts, fundamental security design, encryption, message authentication, malicious software and intruders, and infer how to apply these in general design questions.
- **CLO 3:** Build a software application that implements proper information management and security computing design principles using public and symmetric key cryptography, and role-based access control.
- **CLO 4:** Explain networking and communication, and parallel and distributed computing principles and infer how to apply these in general implementation questions.
- **CLO 5:** Properly implement networking protocols, parallel and distributed algorithms in applications.

Tentative schedule

Week 1	Syllabus, Python Basics
Week 2	Python Basics, programming assignment 1
Week 3	Python Basics, programming assignment 2
Week 4	Parallel Programming with Python
Week 5	Parallel Programming with Python, programming assignment 3
Week 6	Networking and Distributed Programming with Python
Week 7	Networking and Distributed Programming with Python, programming assign. 4
Week 8	Midterm
Week 9	Database and Information Management
Week 10	Database and Information Management, programming assignment 5
Week 11	Security,
Week 12	Security
Week 13	Security, programming assignment 6
Week 14	Security
Week 15	Security

Grading policy

Programming assignments	40%
Midterm exam	20%
Final exam	30%
Homework/Survey/Quizzes/Coding exercises	10%

To earn a C- or better, you must earn at least 60% in the exam component. In other words, to have a C- or better grade, you must earn at least 30 points from midterm and final exams.

Midterm and final will be close book exams, and many questions will be coding questions.

Letter Grade Scale

Letter	Numerical Average
A	92.00 - 100
A-	90.00 - 91.99
B+	88.00 - 89.99
B	82.00 - 87.99
B-	80.00 - 81.99
C+	78.00 - 79.99
C	72.00 - 77.99
C-	69.00 - 71.99
D	62.00 - 68.99
D-	60.00 - 61.99
F	0.00 - 59.99

Incompletes

The grade of 'I' (Incomplete) will be assigned **only** under the following exceptional conditions:

- Due to an extended illness or other extraordinary circumstance, and with appropriate documentation, the student is unable to participate in class for a period affecting graded work, such that it is not possible to complete/grade the missing work before end of course.
- The missing work cannot constitute the majority of the work in the course, and the student must already be passing the course in their other coursework, for an incomplete to be considered
- In such a case, arrangements must be made to make up the missed portion of the course prior to the end of the next semester.

Please note that "Incomplete" is **not** a "get-out-of-bad-grade-and-retake-the-course" card. University policy on Incompletes will be strictly followed.

Exams

A midterm exam and a final exam. The final exam will be cumulative.

Coding questions from programming assignments will account for about 20-30% of the exams.

Programming Assignments

- Tentatively 6 assignments, equally weighted.
 - Some individual, some group of 2 (find your partner).
- Make sure to immediately start working on programming assignments when they are announced. Assignment specifications will be posted on the course web page.
- **Turn in all assignments on time!** Late assignments will be accepted one day after the due date, with the deduction of a letter grade (10%). Assignments more than a day late will not be accepted.
- **Programs that generate ANY kind of errors during execution on linprog (with proper inputs) will receive at most 10% of the total grade for the assignment.** Make sure that your code does not generate any error during execution **on linprog** before you submit it!
- Assignments need to be submitted through canvas. E-mailed files will NOT be counted as valid submissions.
- Any concerns or issues regarding the accuracy of grades must be addressed first to the grader of the assignment in question, and must also be brought up within 7 days of the grade being posted.

ChatGPT, Copilot, etc

- ChatGPT/Copilot are a very powerful and useful tool.
- But in this class, **you are strongly discouraged to use them, but are allowed to.**
 - If you use such tools, you must cite them in your submission.
 - If you use such tools, you must have a good understanding of the generated code. Such code may not have all specified functionality. Additionally, if the code produces any error during execution, it will receive no more than 10% of the points.
 - There will be exam questions on programming assignments. You need to know how to do the assignment to do well in the exam.

Quizzes, homework, etc

- Homework, surveys, and quizzes will be given, both in Canvas short-answer format and in small coding-exercise format. Some quizzes may be set up as auto-graded quizzes on Canvas. Others will be small coding exercises of the type that might be done in a recitation class period (under normal on-campus course delivery conditions). Such coding exercises will be assigned and due in a short time period, always on the same day.

Excused Absence Policy

- Excused absences include documented illness, deaths in the family and other documented crises, call to active military duty or jury duty, religious holy days, and official University activities. These absences will be accommodated in a way that does not arbitrarily penalize students who have a valid excuse. Consideration will also be given to students whose dependent children experience serious illness.
- It is the student's responsibility to notify me as soon as possible regarding any excused absences that affect graded work. In the case of quizzes, you need to notify me within 24 hours of any excused absence that causes you to miss a quiz

Miscellaneous Policies

- The course web page is your friend -- check and refresh it frequently! It will be continually updated with essential course materials, such as assignments, examples, and notes outlines. It will also include other helpful supplements, such as instructions for using the compilers, suggested exercises, and other useful help materials. It is your responsibility to check the web page often for posted materials.
- Students in the class should have a computer account from the Computer Science Department (sign up for one if you don't already have one), and this can and should be used to store project files and access one of the compilers used in the course.
- Knowing the contents of this syllabus is each student's responsibility. "I didn't know" in regards to a clearly stated course policy will NOT be considered a valid excuse for anything during the course

Academic Honor Policy

The Florida State University Academic Honor Policy outlines the University's expectations for the integrity of students' academic work, the procedures for resolving alleged violations of those expectations, and the rights and responsibilities of students and faculty members throughout the process. Students are responsible for reading the Academic Honor Policy and for living up to their pledge to "... be honest and truthful and ... [to] strive for personal and institutional integrity at Florida State University." (Florida State University Academic Honor Policy, found at

<http://fda.fsu.edu/Academics/Academic-Honor-Policy.>)

Academic Honor Policy

- Students are expected to do their **own** work on any classwork or test submitted for a grade (unless designated as a group assignment).
 - It is understandable that discussing a problem with other people may lead to more insight into the issues involved. **While discussing common problems in material or encountered in assignments is fine, discussing or sharing solutions to the problem/assignment is NOT acceptable.** Giving or receiving solutions to/from other students before an assignment is due and handed in (by all parties) will be considered a violation of the Academic Honor Code
 - Discussing solutions and techniques on assignments with other students **after** the assignment has been graded and handed back is okay, and encouraged.
 - When you turn in work with your name on it, you are representing that work as **your own**. If your submission matches that of another student, this is considered a **violation** of the Academic Honor Code.

Academic Honor Policy

- Plagiarism detection tools, such as MOSS, will be used in the grading process.
- Examples found in the course textbook may be used in programs, as long as the source is cited. This is appropriate, as some hand-in assignments may be based on program examples found in the book or contain other code that is provided to you in the assignment specification
- Do NOT post your assignment code solutions to publicly searchable web sites. This includes online compilers/version management systems that may auto-post your code in a publicly searchable way. If you do, somebody else may copy your solutions -- and you'll be on the hook for providing it to them!
- Posting this course's assignments on other online sites/forums in attempts to solicit outside solutions is a violation of the honor code policy AND a violation of copyright. This includes, but is not limited to, web sites like Chegg, Course Hero, Stack Overflow, and others. Copying solutions from these types of sites is clear plagiarism and definitely a violation

Academic Honor Policy

- If it is found that a student has violated the academic honor policy the student is not permitted to drop or withdraw from the course, and must complete the course with the sanctions accessed via the policy. This is a UNIVERSITY policy.
- It is HIGHLY suggested that you only go to our course TA's or the instructor for help if/when needed. Outside tutors are NOT recommended. If you feel you need One on One tutoring, ACE tutoring with FSU is a good resource.
- When two highly similar programs are found, both will be treated as honor code violation. The instructor will not deal with any arguments. They will be resolved at the higher level (university honor court).

Academic Honor Policy

- A first violation of the honor code will result, at **minimum** (but not limited to), a penalty of a 0 grade on the assignment or test involved, along with a reduced letter grade in the course.
- If the violation is that of soliciting solutions on external web sites (Chegg, Course Hero, etc), the penalty will be an automatic F in the course, regardless of whether it is a first violation or not

Academic Honor Policy

- Any second violation of the honor code will result in a Hearing Referral under FSU Honor Code policies (this is an FSU requirement for any case in which a student has a prior violation in ANY course).
- A second violation in THIS course will result in a Hearing Referral with the proposed **minimum** sanction of an F in the course

Americans With Disabilities Act

- Students with disabilities needing academic accommodation should:
(1) register with and provide documentation to the Student Disability Resource Center; and
(2) bring a letter to the instructor indicating the need for accommodation and what type. This should be done during the first week of class.
- This syllabus and other class materials are available in alternative format upon request.

Americans With Disabilities Act

For more information about services available to FSU students with disabilities, contact the:

- **Office of Accessibility Services**

874 Traditions Way

108 Student Services Building

Florida State University

Tallahassee, FL 32306-4167

(850) 644-9566 (voice)

(850) 644-8504 (TDD)

oas@fsu.edu

<https://dsst.fsu.edu/oas>

Syllabus Changes

- Except for changes that substantially affect implementation of the evaluation (grading) statement, this syllabus is a guide for the course and is subject to change with advance notice.

Student Responsibility

- Students should log on to Canvas at least once every two days to check for course updates.
- Students are expected to keep up with the class, engage with the course material, and submit assignments by the due dates.
- Assignments, quizzes, and exams are expected to be products of individual students per the [FSU Academic Honor Policy](#). Students should not discuss the solutions of the questions with each other before or during the actual assignments, activities, quizzes, or exams without instructor approval.
- To receive maximum points for questions, students need to follow the instructions carefully, follow word limits as instructed, and use spell and grammar checking.
- To be successful in this course, students need to complete all required assignments and tests.

Recap

- What language we will be using in this class?
- What you will be learning in this class?
- Is this course all about Python programming?
- How many exams will there be?
- (True or False) I will pass the class if I score 100% in programming assignments and 50% in the exams?

Recap

- How many partial points can you get in a programming assignment if your program generates error during execution on linprog?
- When should you start your programming assignment?
- Are you allowed to use ChatGPT in this class?
- Can you seek help in Chegg?
- What is the penalty for the first violation of the honor code?